D06 · Logistics Governance & Analytics

05 · Reverse logistics & returns management

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Reverse logistics manages products moving back from customers — returns, repairs, recalls, end-of-life — from authorization and collection through inspection and disposition: resell, refurbish, liquidate or scrap.

2 WHY IT MATTERS

Returns are often the least mature process in the network, and in e-commerce they can be large enough to erase the margin of the original sale. **Value decays with time, so slow disposition destroys recoverable value (Rogers and Tibben-Lembke — Going Backwards: Reverse Logistics Trends and Practices)**. Returns data is also the best source of information for reducing future returns.

3 HOW IT IS DONE

1. **Gate returns at authorization.** Reason codes and rules at RMA to filter and route returns before they move.
2. **Define disposition rules in advance.** Clear criteria for resell, refurbish, liquidate or scrap by category and condition.
3. **Process fast.** Measure receipt-to-disposition time and treat it as a value metric.
4. **Feed root causes back.** Share reasons with quality, packaging, product content and sales every month.

IN PRACTICE

Define disposition criteria before the returns arrive

Ad-hoc decisions at the moment of inspection are slow and inconsistent. Pre-defined rules accelerate flow and protect value.

Speed of disposition is itself a value lever

Every day a return sits unprocessed reduces the probability and value of successful resale or refurbishment.

Close the loop to the teams that can reduce future returns

Quality, packaging, product description and commercial promise accuracy all influence return rates. Reverse logistics data is wasted if it never reaches them.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — E-commerce operation with 22% return rate and manual disposition

Metric	Before	After Process Redesign
Average time from receipt to disposition	8.4 days	2.1 days
Value recovered per $ of returned product	$0.38	$0.62
Disposition criteria	Ad-hoc	Documented 4-path rules
Root-cause feedback to commercial / quality	Almost none	Monthly structured review

Result: Recovery rate improved 63%. More importantly, systematic root-cause data allowed packaging and product-description changes that began to reduce the return rate itself.

HOW IT IS MEASURED

Return Rate %

(Units returned / Units sold) × 100

Benchmark: <3% industrial B2B · <10% physical retail · <25% fashion e-commerce (category dependent)

Return Recovery Rate %

(Value recovered from disposed returns / Book value of those returns) × 100

Benchmark: >60% well-managed e-commerce · >80% high-value B2B

WHAT YOU WOULD USE

Reverse Logistics

Optoro / Loop / specialized returns platforms — Strong for high-volume e-commerce returns management, RMA and disposition optimization.

SAP S/4HANA Returns Management / WMS returns flows — Integrated reverse flows for companies already on major ERP/WMS platforms.

4 SO WHAT

Run returns as a process, not an afterthought. Gate them at authorization, decide disposition by rule, process fast and send the reasons back to the teams that can prevent them.

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